

AJEEM.S

AI/ML Engineer & AI Developer | B.Tech Artificial Intelligence & Data Science

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B.Tech AI & Data Science • 2024–2028

PROFILE

AI/ML developer focused on practical AI systems, Tamil NLP, RAG, OCR, local-first LLM applications, voice interfaces, and full-stack AI engineering. Builds production-oriented systems combining FastAPI, React, vector search, LLMs, automation, and local inference.

PROJECT PORTFOLIO

TAMIL-AI.ME

- Built a Tamil-first NLP/RAG platform with FastAPI, React, ChromaDB, semantic retrieval, and AI-assisted document understanding for Tamil-focused knowledge applications.
- Curated 4,500+ literary entries across 22 sources, including Thirukkural and Sangam literature, enabling semantic search and multilingual retrieval.
- Integrated IndicTrans2 for Tamil↔English translation and KalVettu OCR using TrOCR + OpenCV for extracting and interpreting Chola-era Tamil stone inscriptions.

SECRAIN

- Designed a local-first AI memory and reasoning engine using FastAPI, SQLAlchemy, SQLite/ChromaDB, and Ollama for private storage and retrieval of conversations, files, notes, and project data.
- Implemented automatic memory extraction, semantic search, structured personal/project data, notes retrieval, and related-data mapping to make stored information usable by an AI assistant.
- Built hybrid inference support that can route workloads between cloud LLM APIs and local Ollama models, providing an offline-capable foundation for private AI workflows.

TITAN — Personal AI Assistant

- Developed a desktop AI assistant with React/Vite, FastAPI, voice interaction, task management, live system monitoring, browser/app automation, and a plugin-oriented architecture.
- Implemented responsive assistant communication using WebSocket + REST patterns, with system-control workflows for opening applications, folders, files, websites, and browser tabs.
- Built voice pipeline components around a custom wake-word workflow, faster-whisper speech recognition, WebRTC VAD, and local/cloud AI routing for Tamil, English, and Tanglish interactions.

ROGERS — AI Student Project Orchestrator

- Designed an AI-driven workflow that transforms a student's one-line project idea into a structured software project through research, refinement, model selection, planning, implementation, testing, and delivery.
- Built the current application foundation with FastAPI/Pydantic, Uvicorn, React, Vite, TypeScript, Tailwind CSS, Lucide, and an offline BuiltinProvider alongside cloud-model integration.
- Designed a multi-model orchestration approach in which specialized AI agents can research, generate backend/frontend code, test, debug, and package a completed project with user approval at key stages.

AIRES — AI Emergency Response Platform

- Designed an AI-assisted emergency response workflow covering call intake, incident understanding, intelligent dispatch, route planning, and hospital recommendation.
- Structured the system around real-time event handling so emergency resources and routing decisions can be updated as simulated incident conditions change.
- Focused the project on combining AI decision support with practical emergency-service workflows rather than treating the model as a standalone chatbot.

ADDITIONAL AI PROJECTS

VCH-AGENT — Local-First Content Intelligence

- Developed a local-first AI workflow concept for analyzing Reddit/YouTube content and identifying signals associated with potential virality using lightweight local models and structured data.
- Designed the system around Python, SQLite/Gradio, and local inference to minimize API cost while keeping the workflow suitable for experimentation and iterative model evaluation.
- Focused the project on turning raw social/video signals into actionable content insights that can support creator workflows and platform-specific customization.

NEWS-BOT — Automated News Digest

- Designed an automated Telegram news-digest workflow that collects information from RSS feeds and organizes it into a scheduled morning briefing.
- Structured the pipeline around category-based collection and LLM summarization, with a target of delivering a concise digest around 8:00 AM IST.
- Planned the system with Gemini 2.5 Flash and multiple RSS categories to reduce manual news monitoring while keeping the workflow source-driven.

EYE-CARE TOOL

- Developed an AI/computer-vision project focused on practical eye-care assistance and user-facing analysis rather than a general-purpose chatbot.
- Organized the project as a standalone local application so experimentation, model integration, and testing can be performed directly on the development machine.
- Applied a local-first engineering approach with lightweight processing, a clear user workflow, and separation between AI inference and application logic.

CivicPulse

- Designed a civic AI assistant concept for triaging civic questions and directing users toward appropriate information or next-step workflows.
- Planned multilingual interaction across English, Tamil, and Tanglish to make civic information more accessible to users who do not prefer formal English interfaces.
- Focused the architecture on classification/triage and information routing rather than allowing the model to invent authoritative civic answers.

TECHNICAL SKILLS

Programming:	Python, TypeScript, JavaScript, SQL, HTML/CSS
AI / ML:	NLP, RAG, Hugging Face Transformers, IndicTrans2, Ollama, LoRA/Fine-tuning, embeddings
Backend / Frontend:	FastAPI, REST APIs, React, Vite, Tailwind CSS, WebSocket
Data / Retrieval:	SQLite, PostgreSQL, ChromaDB, vector search, semantic search, BM25
AI Systems / Tools:	OpenCV, TrOCR, faster-whisper, WebRTC VAD, Git, GitHub, Docker

EXPERIENCE

KaaShiv Infotech — Data Science Internship | Grade A

IBM SkillsBuild AICTE Internship — via Edunet Foundation | Ongoing

CERTIFICATIONS

- Oracle Fusion AI Agent Studio — Certified Foundations Associate
- Tata Forage — GenAI Data Analytics
- Claude MCP Certificate

EDUCATION

B.Tech Artificial Intelligence and Data Science — Dhanalakshmi Srinivasan University, Perambalur, Tamil Nadu | 2024–2028

Higher Secondary (12th) — St. Joseph Academy, Sankarapuram | 74%

Secondary School (10th) — St. Joseph Academy, Sankarapuram | 88%

CORE STRENGTHS

AI application development • RAG & semantic retrieval • Tamil NLP • Local-first AI • Full-stack development • AI system integration
• Problem solving • AI automation • Voice interfaces

PROFESSIONAL LINKS

GitHub — github.com/ajeem-suban

LinkedIn — linkedin.com/in/ajeem-suban

Portfolio — ajeem-suban.vercel.app